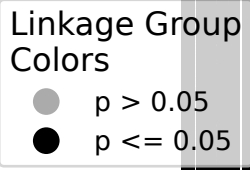


diamond: 14190 orthologs on 18 Mare scaffolds and 5053 Rphi scaffolds.



d Mare vs Rphi FET, whole chroms.

	Mare scaf	Rphi scaf	p (FET)	Count
0	NC_069133.1	NW_026851633.1	6.93e-25	24
1	NC_069132.1	NW_026861563.1	1.01e-19	23
2	NC_069125.1	NW_026861840.1	1.42e-15	23
3	NC_069132.1	NW_026865484.1	1.22e-18	22
4	NC_069132.1	NW_026861019.1	1.22e-18	22
5	NC_069123.1	NW_026862663.1	5.77e-19	22
6	NC_069123.1	NW_026860903.1	7.74e-17	21
7	NC_069123.1	NW_026852705.1	6.60e-15	20
8	NC_069132.1	NW_026865571.1	3.38e-12	19
9	NC_069136.1	NW_026863645.1	1.88e-18	18
10	NC_069123.1	NW_026852008.1	1.42e-14	18
11	NC_069135.1	NW_026858503.1	2.07e-16	17
12	NC_069134.1	NW_026859933.1	5.27e-14	17
13	NC_069129.1	NW_026863896.1	1.26e-15	17
14	NC_069129.1	NW_026857770.1	1.18e-12	17
15	NC_069125.1	NW_026859561.1	6.03e-11	17
16	NC_069125.1	NW_026855197.1	1.98e-07	17
17	NC_069124.1	NW_026852258.1	2.05e-09	17
18	NC_069123.1	NW_026853170.1	1.05e-14	17
19	NC_069134.1	NW_026865967.1	4.41e-14	16
20	NC_069134.1	NW_026864352.1	6.99e-13	16
21	NC_069132.1	NW_026863541.1	2.38e-13	16
22	NC_069129.1	NW_026855294.1	6.98e-11	16
23	NC_069134.1	NW_026854960.1	6.20e-13	15
24	NC_069126.1	NW_026860085.1	1.10e-12	15
25	NC_069125.1	NW_026857221.1	1.89e-07	15
26	NC_069123.1	NW_026862783.1	2.68e-11	15
27	NC_069135.1	NW_026863406.1	1.32e-11	14
28	NC_069134.1	NW_026860451.1	1.22e-10	14
29	NC_069130.1	NW_026867184.1	7.71e-13	14
30	NC_069129.1	NW_026861864.1	5.78e-11	14
31	NC_069129.1	NW_026861084.1	2.30e-09	14
32	NC_069129.1	NW_026857630.1	4.11e-12	14
33	NC_069127.1	NW_026862474.1	1.26e-10	14
34	NC_069127.1	NW_026852998.1	1.19e-12	14
35	NC_069123.1	NW_026867092.1	3.28e-10	14
36	NC_069123.1	NW_026856555.1	2.35e-11	14
37	NC_069123.1	NW_026855648.1	3.28e-10	14
38	NC_069123.1	NW_026853802.1	2.44e-09	14
39	NC_069123.1	NW_026852524.1	2.35e-11	14
40	NC_069138.1	NW_026852000.1	1.66e-12	13
41	NC_069135.1	NW_026863691.1	2.03e-10	13
42	NC_069133.1	NW_026866838.1	3.02e-12	13
43	NC_069132.1	NW_026862689.1	6.21e-09	13
44	NC_069132.1	NW_026857788.1	6.21e-09	13
45	NC_069132.1	NW_026853905.1	4.32e-08	13
46	NC_069129.1	NW_026860420.1	6.08e-11	13
47	NC_069125.1	NW_026860502.1	2.37e-03	13
48	NC_069124.1	NW_026866929.1	1.69e-06	13
49	NC_069124.1	NW_026866221.1	1.01e-04	13
50	NC_069124.1	NW_026865813.1	2.50e-07	13
51	NC_069124.1	NW_026861602.1	1.98e-08	13
52	NC_069124.1	NW_026857083.1	1.98e-08	13
53	NC_069138.1	NW_026863843.1	3.69e-11	12
54	NC_069135.1	NW_026867313.1	3.10e-09	12
55	NC_069134.1	NW_026856260.1	1.36e-07	12
56	NC_069132.1	NW_026866837.1	7.27e-08	12
57	NC_069132.1	NW_026862956.1	7.27e-08	12
58	NC_069132.1	NW_026859620.1	7.27e-08	12
59	NC_069132.1	NW_026852468.1	6.04e-09	12
60	NC_069129.1	NW_026863615.1	9.00e-10	12
61	NC_069129.1	NW_026861130.1	3.37e-07	12
62	NC_069129.1	NW_026855204.1	1.10e-08	12
63	NC_069127.1	NW_026864590.1	3.10e-10	12

e

Mare scaffolds

Mare vs Rphi FET, chrom. pieces

	Mare scaf	Rphi scaf	p (FET)	Count
0	NC_069133.1 : 0-end	NW_026851633.1 : 0-end	6.93e-25	24
1	NC_069132.1 : 0-end	NW_026861563.1 : 0-end	1.01e-19	23
2	NC_069125.1 : 0-end	NW_026861840.1 : 0-end	1.42e-15	23
3	NC_069132.1 : 0-end	NW_026865484.1 : 0-end	1.22e-18	22
4	NC_069132.1 : 0-end	NW_026861019.1 : 0-end	1.22e-18	22
5	NC_069123.1 : 0-end	NW_026862663.1 : 0-end	5.77e-19	22
6	NC_069123.1 : 0-end	NW_026860903.1 : 0-end	7.74e-17	21
7	NC_069123.1 : 0-end	NW_026852705.1 : 0-end	6.60e-15	20
8	NC_069132.1 : 0-end	NW_026865571.1 : 0-end	3.38e-12	19
9	NC_069136.1 : 0-end	NW_026863645.1 : 0-end	1.88e-18	18
10	NC_069123.1 : 0-end	NW_026852008.1 : 0-end	1.42e-14	18
11	NC_069135.1 : 0-end	NW_026858503.1 : 0-end	2.07e-16	17
12	NC_069134.1 : 0-end	NW_026859933.1 : 0-end	5.27e-14	17
13	NC_069129.1 : 0-end	NW_026863896.1 : 0-end	1.26e-15	17
14	NC_069129.1 : 0-end	NW_026857770.1 : 0-end	1.18e-12	17
15	NC_069125.1 : 0-end	NW_026859561.1 : 0-end	6.03e-11	17
16	NC_069125.1 : 0-end	NW_026855197.1 : 0-end	1.98e-07	17
17	NC_069124.1 : 0-end	NW_026852258.1 : 0-end	2.05e-09	17
18	NC_069123.1 : 0-end	NW_026853170.1 : 0-end	1.05e-14	17
19	NC_069134.1 : 0-end	NW_026865967.1 : 0-end	4.41e-14	16
20	NC_069134.1 : 0-end	NW_026864352.1 : 0-end	6.99e-13	16
21	NC_069132.1 : 0-end	NW_026863541.1 : 0-end	2.38e-13	16
22	NC_069129.1 : 0-end	NW_026855294.1 : 0-end	6.98e-11	16
23	NC_069134.1 : 0-end	NW_026854960.1 : 0-end	6.20e-13	15
24	NC_069126.1 : 0-end	NW_026860085.1 : 0-end	1.10e-12	15
25	NC_069125.1 : 0-end	NW_026857221.1 : 0-end	1.89e-07	15
26	NC_069123.1 : 0-end	NW_026862783.1 : 0-end	2.68e-11	15
27	NC_069135.1 : 0-end	NW_026863406.1 : 0-end	1.32e-11	14
28	NC_069134.1 : 0-end	NW_026860451.1 : 0-end	1.22e-10	14
29	NC_069130.1 : 0-end	NW_026867184.1 : 0-end	7.71e-13	14
30	NC_069129.1 : 0-end	NW_026861864.1 : 0-end	5.78e-11	14
31	NC_069129.1 : 0-end	NW_026861084.1 : 0-end	2.30e-09	14
32	NC_069129.1 : 0-end	NW_026857630.1 : 0-end	4.11e-12	14
33	NC_069127.1 : 0-end	NW_026862474.1 : 0-end	1.26e-10	14
34	NC_069127.1 : 0-end	NW_026852998.1 : 0-end	1.19e-12	14
35	NC_069123.1 : 0-end	NW_026867092.1 : 0-end	3.28e-10	14
36	NC_069123.1 : 0-end	NW_026856555.1 : 0-end	2.35e-11	14
37	NC_069123.1 : 0-end	NW_026855648.1 : 0-end	3.28e-10	14
38	NC_069123.1 : 0-end	NW_026853802.1 : 0-end	2.44e-09	14
39	NC_069123.1 : 0-end	NW_026852524.1 : 0-end	2.35e-11	14
40	NC_069138.1 : 0-end	NW_026852000.1 : 0-end	1.66e-12	13
41	NC_069135.1 : 0-end	NW_026863691.1 : 0-end	2.03e-10	13
42	NC_069133.1 : 0-end	NW_026866838.1 : 0-end	3.02e-12	13
43	NC_069132.1 : 0-end	NW_026862689.1 : 0-end	6.21e-09	13
44	NC_069132.1 : 0-end	NW_026857788.1 : 0-end	6.21e-07	13
45	NC_069132.1 : 0-end	NW_026853905.1 : 0-end	4.32e-08	13
46	NC_069129.1 : 0-end	NW_026860420.1 : 0-end	6.08e-11	13
47	NC_069125.1 : 0-end	NW_026860502.1 : 0-end	2.37e-03	13
48	NC_069124.1 : 0-end	NW_026866929.1 : 0-end	1.69e-06	13
49	NC_069124.1 : 0-end	NW_026866221.1 : 0-end	1.01e-04	13
50	NC_069124.1 : 0-end	NW_026865813.1 : 0-end	2.50e-07	13
51	NC_069124.1 : 0-end	NW_026861602.1 : 0-end	1.98e-08	13
52	NC_069124.1 : 0-end	NW_026857083.1 : 0-end	1.98e-08	13
53	NC_069138.1 : 0-end	NW_026863843.1 : 0-end	3.69e-11	12
54	NC_069135.1 : 0-end	NW_026867313.1 : 0-end	3.10e-09	12
55	NC_069134.1 : 0-end	NW_026856260.1 : 0-end	1.36e-07	12
56	NC_069132.1 : 0-end	NW_026866837.1 : 0-end	7.27e-08	12
57	NC_069132.1 : 0-end	NW_026862956.1 : 0-end	7.27e-08	12
58	NC_069132.1 : 0-end	NW_026859620.1 : 0-end	7.27e-08	12
59	NC_069132.1 : 0-end	NW_026852468.1 : 0-end	6.04e-09	12
60	NC_069129.1 : 0-end	NW_026863615.1 : 0-end	9.00e-10	12
61	NC_069129.1 : 0-end	NW_026861130.1 : 0-end	3.37e-07	12
62	NC_069129.1 : 0-end	NW_026855204.1 : 0-end	1.10e-08	12
63	NC_069127.1 : 0-end	NW_026864590.1 : 0-end	3.10e-10	12

(a) depicts the chromosomes/scaffolds of Mare (x-axis) plotted against the chromosomes/scaffolds of Rphi (y-axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 14190 orthologs found between 18 Mare scaffolds and 5053 Rphi scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. *Nature Ecology & Evolution* 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes. (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chromosome FET p-value, and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not show the correlation matrix. (e) shows the FET p-value of the correlation between chromosome pairs. Useful for showing if single arms of chromosomes are correlated with other regions.